

The Dataverse Network: An Infrastructure for Data Sharing

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(5/28/08 talk at the Society for Scholarly Publishing)

- Gary King, **An Introduction to the Dataverse Network as an Infrastructure for Data Sharing**, *Sociological Methods and Research*, 32, 2 (November, 2007): 173–199.
- Micah Altman and Gary King. **A Proposed Standard for the Scholarly Citation of Quantitative Data**, *D-Lib Magazine*, 13, 3/4 (March/April).
- Dataverse Network project: <http://TheData.org>

- Accessibility:
 - Most large data sets: in public archives
 - Most data in published articles: not accessible, results not replicable without the original author
 - Most data sets from NSF & NIH grants: not publicly available
- Problems even with professional archives:
 - Data in different archives have different identifiers
 - One major archive renumbered all its acquisitions
 - Changes to data are made; identifiers are reused or deaccessioned; old data are lost
- Data sets are not like books
 - Static data files (even if on the web): unreadable after a few years
 - When storage methods change: some data sets are lost; others have altered content!

What About a Centralized Data Access Solution?

- Highly desirable when feasible
- Works great in genomics, astronomy, etc., when data formats are universal, goals are common, and agreements are in place
- **Impossible** when data are heterogeneous in format, origin, size, effort needed to collect or analyze, IRB access rules, etc.
- Why don't researchers put data in public archives?
 - The Archive gets the credit
 - Upon questioning: they want credit, control, and visibility
 - (So why don't they worry about print publishers getting all the credit? Lack of data citations!)
- We propose: technological solutions to these political and social problems

Requirements for Effective Data Sharing Infrastructure

- **Recognition**, for authors, journals, etc. in (1) citations to data, (2) citations to associated articles, and (3) visibility on the web.
- **Public Distribution**, without permission from the author
- **Authorization**: fulfill requirements the author originally met
- **Validation**: check that data exists, without authorization
- **Persistence** Decades from now. . . .
- **Verification**: data remains unchanged, even if converted from SPSS to Stata to R, from a PC to a Mac to Linux, and from 8 inch magnetic tape to 5.25 inch floppies to a DVD.
- **Ease of Use** Neither editors nor authors employ professional archivists
- **Legal Protection**:
 - Journals have liability protection for print; none for data
 - *If you put data on the web without IRB approval, you are violating federal regulations*
 - (IRB approval must be for data distribution, not merely for the study)
 - Solution must not require lawyers (we've automated the IRB)

Rules for Citing Printed Matter

Kim, Jae-On, Norman Nie, and Sidney Verba. 1977. "A Note on Factor Analyzing Dichotomous Variables: The Case of Political Participation," Political Methodology, Vol. 4: No. 2 (Spring): Pp. 39–62.

First author (last name first) Second author Third author Year Article title Journal (no longer exists) Volume number Issue number Season Pages Special formatting codes Special indentation Citations: rule-based, precise, redundant Print Citations Work: authors don't think publishers get all the credit; cited articles can be found; copyeditors don't need to see the original to know it exists; the link from citation to print persists

A New Citation Standard for Numeric Data

Sidney Verba, 1998, "Political Participation Data", [hdl:1902.4/00754](https://hdl.handle.net/1902.4/00754),
UNF:3:6:ZNQRI14053UZq389x0Bffg?== Annals of Applied Statistics
[Distributor]; NORC [Producer].

- 1 Author
- 2 Year
- 3 Title
- 4 Unique Global Identifier: will work after URLs stop working
- 5 Linked to a Bridge Service (presently a URL:
<http://id.thedata.org/hdl%3A1902.4%2F00754>)
- 6 Universal Numeric Fingerprint (UNF)
- 7 Standard rules for adding citation elements

Data to Universal Numeric Fingerprints

$$\begin{pmatrix} 1 & 4 & 4 & 21 & \dots & 121 \\ 1 & 2 & 2 & 91 & \dots & 212 \\ 1 & 9 & 2 & 72 & \dots & 104 \\ 0 & 2 & 2 & 2 & \dots & 321 \\ 1 & 6 & 2 & 12 & \dots & 204 \\ 1 & 9 & 4 & 52 & \dots & 311 \\ 0 & 3 & 2 & 23 & \dots & 92 \\ 0 & 2 & 5 & 91 & \dots & 212 \\ 0 & 5 & 8 & 91 & \dots & 91 \\ 1 & 9 & 1 & 72 & \dots & 104 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 2 & 91 & \dots & 212 \end{pmatrix} \Rightarrow \text{ZNQRI14053UZq389x0Bffg?}==$$

Advantages of UNFs

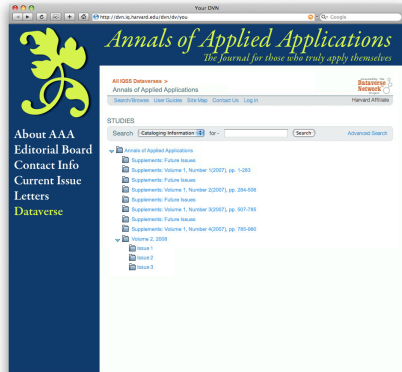
- UNF is **calculated from the content** not the file: Its the **Same UNF** regardless of changes in computer hardware, storage medium, operating system, statistical software, database, or spreadsheet software.
- Cryptographic technology: **any** change in data content changes the UNF. (cannot tinker after the fact!)
- Noninvertible properties
 - UNFs convey no information about data content
 - OK to distribute for highly sensitive, confidential, or proprietary data
 - Copyeditor can validate data's existence even without authorization
- The citation refers to **one specific data set** that can't **ever** be altered, even if journal doesn't keep a copy
- Future researchers can quickly check that they have the same data as used by the author: merely recalculate the UNF

Web 2.0 Terminology

- **Software**: find CD, install locally, hit next, hit next, hit next. . .
- **Web application software**: no installation; load web browser and run (Dataverse Network Software)
- **Host**: The computers where the web application software runs (universities, archives, libraries)
- **Virtual host**: Where the web application software *seems* to run, but does not (web sites of: authors, journals, granting agencies, research centers, universities, scholarly organizations, etc.)



Your web site



Your dataverse branded as your web site but served by the Dataverse Network, therefore re-quiring no local installation and providing an enormous array of services

http://dm.is.harvard.edu/dm/dv/you

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REPLICATION DATA FOR: IMPROVING FORECASTS OF STATE FAILURE

Cataloging Information Documentation, Data and Analysis

Citation Information

Gary King, Langhe Zeng, 2001, "Replication data for Improving Forecasts of State Failure", hdl:1902.1/RPQDQWNR
UNF:3:CbafGhbaDyHdYhWwWQWw Murrey Research Archive [Distribution]

How to Cite

Study Global Id hdl:1902.1/RPQDQWNR

Authors Gary King, Langhe Zeng

Production Date 2001

Distributor Murrey Research Archive **MRA**

Distributor Contact mrx_support@help.mra.harvard.edu

Deposit Date 2006

Replication For King, Gary, Zeng, Langhe, 2001, "Improving Forecasts of State Failure," World Politics, Vol. 53, No. 4, 623-66. http://king.harvard.edu/teaching/private-sites/psm/article_evaluation.html.

Provenance Gary King Dataserve

Abstract and Scope

We offer the first independent scholarly evaluation of the claims, forecasts, and causal inferences of the State Failure Task Force and their efforts to forecast when states will fail. State Failure refers to the collapse of the authority of the central government to impose order, as in civil wars, revolutionary wars,

http://dm.is.harvard.edu/dm/dv/you

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REPLICATION DATA FOR: IMPROVING FORECASTS OF STATE FAILURE

Cataloging Information Documentation, Data and Analysis

Download all files in a single archive file (files that you cannot access will not be downloaded)

File Name	Description	Cases/ Variables	Type	Controls
1. Documentation				
improvingforecasts.pdf	Article related to this study: Improving Forecasts of State Failure		application/pdf	
2. Replication Data				
ns0.spc	Sample spec file for committee n/s			
ns.dat	For use with committee n/s, ASCII data file			
ns.tab	For use with committee n/s	7190 11	Tab delimited	
replication.txt	Additional document describing the format of the replication files			
3. Recoded Data				
names.dta	Country names, in original state format			
names.tab	Country Names Data	8980 3	Tab delimited	
sort111.dta	Recoded Full Data in Original Format, comma delimited text file			
4. Original Data				
statefailure.txt	Documentation for Original		replication data	

http://dvn.harvard.edu/dvn/dvnyou

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REPLICATION DATA FOR: CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE
DATA FILE: CONSTITUENCYSERVICE.TAB [Back to Study](#)

[Download Subset](#) [Subset and Recode](#) [Descriptive Statistics](#) [Advanced Statistical Analysis](#)

Selected Variables

To recode (subset) a variable into a different one, first, select a variable from the selected-variables box, click the arrow button below, and then the name and label of the variable you have chosen appear in the new-variable name and label boxes for convenience. You must replace the old variable name with a unique variable name that is not used in the data file; the new variable label is optional and you can leave it blank.

New Variable Name:
New Variable Label:

[Apply Recodes](#)

Select variables from table below (selected variables will be displayed above) You do not have access to the recoding and analysis functionality for this restricted data file. You can only view the variables and their summary statistics.

Show 20 Variables

Variable Type	Variable Name/Variable Label	Quick Summary
☐ Continuous	INCAD	incumbency advantage
☐ Continuous	BUD	Budget figures
☐ Continuous	SAL	salary
☐ Discrete	CO	state level dummy variables

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REPLICATION DATA FOR CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE
DATA FILE: CONSTITUENCYSERVICE.TAB

Download Subset Subset and Recode Descriptive Statistics **Advanced Statistical Analysis**

Selected Variables

Choose a Statistical Model

Categorical Data Analysis
Cross-Tabulation

Ecological Inference models
Hierarchical Multinomial-Dirichlet Ecological Inference Model for R x C Tables

Select Event Count Models

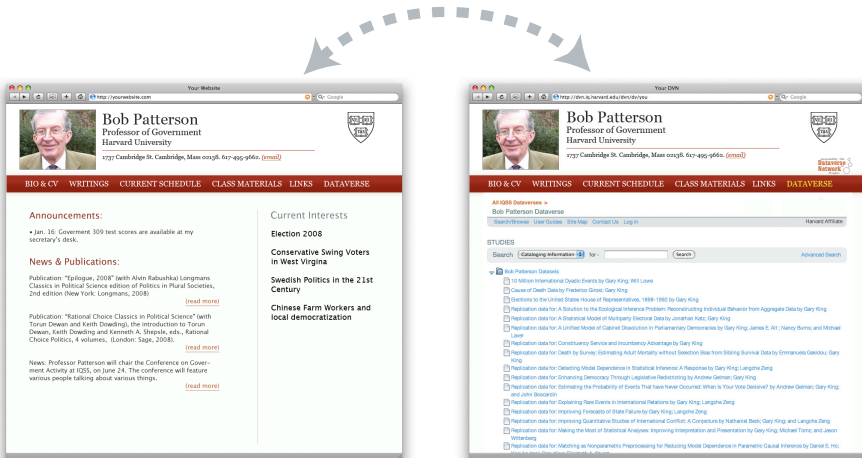
Variables all

- Negative Binomial Regression for Event Count Dependent Variables
- Poisson Regression for Event Count Dependent Variables
- Generalized Additive Model for Event Count Dependent Variables
- General Estimating Equation for Poisson Regression
- Social Network Poisson Regression for Event Count Dependent Variables

Models for Continuous Bounded Dependent Variables

- Cox Proportional Hazard Regression for Duration Dependent Variables
- Exponential Regression for Duration Dependent Variables
- Gamma Regression for Continuous, Positive Dependent Variables
- General Estimating Equation for Gamma Regression

Variable	Model	Link
Discrete	CF	state level dummy variables
Discrete	DE	state level dummy variables
Discrete	JA	state level dummy variables
Discrete	NI	state level dummy variables
Discrete	NO	state level dummy variables
Continuous	SAL	salary
Discrete	GO	2500 level dummy variables



Your web site

Your dataverse branded as your web site but served by the Dataverse Network, therefore requiring no local installation and providing an enormous array of services

The Dataverse Network Project Homepage (<http://TheData.org>)

Dataverse Networks may harvest metadata from each other (dashed lines)

The IQSS Dataverse Network
at Harvard University

SMR Dataverse

Gary King's Dataverse

PoliSci 101 Dataverse

U.S. Census Dataverse

APSA Legislative Section
Dataverse

NSF Dataverse

Weatherhead Center
Dataverse

MacArthur Network on
Inequality Dataverse

Dept of Psychology
Dataverse

[Other] Dataverse Network[s]

Each Dataverse Network may serve numerous individual dataverses (arrows)

Most users come directly to a dataverse, which is a self-contained archive (with all services provided by a Dataverse Network)

Dataverse owners may list, or allow searching across, other dataverses or the data sets in them

Software is available at the project homepage and only needs to be installed to establish a Dataverse Network. Dataverses are virtual hosts.

A Journal Dataverse for a Replication Data Archive

- **Full service virtual archive**, with numerous data services (citation, metadata, archiving, subsetting, conversion, translation, analysis, ...)
- **Hierarchical list of data**: year/volume/issue/article/dataset
- **Branded as yours**: with the look and feel of your site
- **Easy to setup**: give DVN your style, and include a link to your new dataverse
- **Easy to manage**: no software or hardware installation, backups, worry about archiving standards, or data format transations; still exists if you move; easy to rebrand
- **Does not disrupt workflow**: copyeditor ensures data is cited; author is given password to upload data. Everything else is self-service
- **High acceptability**: experiments indicate $> 90\%$ uptake for authors (without publication at stake)
- **Reuse**: data may appear on author's dataverse too
- **Results**: Journals with replication policies have three times the impact factor! (with dataverse, it should be more)

Dataverse Uses

- Journals, for replication data archives
- Authors, for their own data
- Future Researchers: browse or search for a dataverse or dataset; forward citation search; verification via UNFs; subsetting; read metadata, abstract, & documentation; check for new versions; translate format; statistical analyses; download
- Teachers, a list or for in depth analysis
- Sections of scholarly organizations, to organize existing data
- Granting agencies
- Research centers
- Major Research Projects
- Academic departments, universities, data centers, libraries
- Data archives
- Using DVN tools with outside data

The Universe of Data meets the Universe of Methods

- **R Project for Statistical Computing**

- nearly 1000 packages; most new methods appear in R first
- Highly diverse examples, syntax, documentation, and quality
- Difficult for statistics-types; harder for applied researchers

- **Zelig: Everyone's Statistical Software**

- An ontology we developed of almost all statistical methods
- Users incorporate original packages via simple bridge functions via a simple model description language
- Result: Unified Syntax, the same 3 commands to use any method
- Automatically generated Graphical User Interface with all the world's methods

- **R + Zelig + Dataverse Network**

- Greatly reduced time from methods development to use
- Easy for applied researchers even if non-programmers
- Can save R script for replication or further analysis

- **Web application software**
 - Pros: easy to use, no installation costs
 - Cons: the software can vanish or change at any time
- **Dataverse Network Software**
 - Affero GPL License
 - Open source, public ownership
 - If you don't like the new version: you can make a new one
 - You own the software & the underlying code
 - The license guarantees that this will remain true in the future
- **Licensing data**
 - DVN automates the IRB process; no lawyers necessary
 - Data may be restricted in many ways, while metadata is available

For more information

<http://TheData.org>

Technology used in DVN Software

- Language: **Java Enterprise Edition 5** (with EJB3 and JSF) (team picked for JavaOne; Sun engineers regularly call for advice)
- Application server: **GlassFish** (wrote press release on our project)
- Database: we use **PostgreSQL** (can substitute others)
- Statistical computing: **R** and **Zelig**